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**UNIVERSITY OF BATANGAS – LIPA CAMPUS
COLLEGE OF ENGINEERING AND ARCHITECTURE
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COACH AI-rinx

An Agentic AI-Driven Voice Workout Coaching and Tracking
System

Final Examination

Feedback and Control Systems

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Introduction

Coach AI-rinx is an AI-powered voice workout coaching and analytics system designed to help users track workouts more efficiently using Telegram, automation workflows, and artificial intelligence. The system allows users to send workout information through voice messages or text messages, and the AI automatically converts the workout into structured data such as exercise name, weight, repetitions, sets, workout category, and total volume.

The project uses Telegram as the main communication platform, n8n as the workflow automation system, Python Flask with faster-whisper for speech-to-text transcription, Google Sheets as the workout database, and Gemini 3.5 Flash for intelligent AI coaching and workout analysis.

The purpose of Coach AI-rinx is to simplify workout logging and provide users with instant AI-generated workout coaching and progress tracking. Instead of manually typing workout logs into fitness applications, the user can simply send a voice message such as:

“Inclined 24 kilogram 8 reps 3 sets.”

The system automatically transcribes the voice message, extracts workout data, calculates workout analytics, stores the information into Google Sheets, detects personal records, and sends motivational AI coaching responses back through Telegram.

The project was created to demonstrate how artificial intelligence, speech recognition, workflow automation, and data analytics can be combined into a practical real-world application. The system helps users monitor workout progress, reduce manual logging effort, and receive intelligent coaching feedback using modern AI technologies.

Objectives

General Objective

The main objective of this project is to develop an AI-powered voice workout coaching system using workflow automation and artificial intelligence technologies.

Specific Objectives

This project aims to:

- Create a Telegram-based AI workout assistant.
- Implement speech-to-text transcription using Whisper AI.
- Automatically extract workout information from voice messages.
- Calculate workout analytics such as total volume.
- Categorize exercises automatically.



- Store workout logs using Google Sheets.
- Detect personal records (PRs) automatically.
- Generate AI coaching responses using Gemini 3.5 Flash.
- Produce workout summaries and progress analysis.
- Demonstrate hybrid AI system architecture using deterministic logic and AI reasoning.

Equipment and Materials

The following equipment, software, and technologies are needed in making this project:

- Personal Computer or Laptop
- Stable Internet Connection
- Telegram Application
- n8n Workflow Automation
- Python
- Flask Framework
- faster-whisper AI Model
- Google Sheets
- Gemini 3.5 Flash API
- ngrok
- Visual Studio Code

Procedure

1. Telegram Bot Setup

- Create a Telegram bot using BotFather.
- Configure the Telegram bot token inside n8n.
- Enable Telegram Trigger nodes for receiving user messages.

2. Python Whisper AI Server Setup

- Install Python and required libraries.
- Install Flask and faster-whisper packages.
- Create a Flask server with a POST /transcribe endpoint.
- Configure the server to receive Telegram audio files.
- Run the local Flask server.

Example command:
`python server.py`

3. n8n Workflow Setup

- Create Telegram Trigger nodes.
- Create HTTP Request nodes for downloading Telegram voice files.
- Connect the workflow to the Flask Whisper server.
- Add JavaScript nodes for parsing workout information.



- Add category detection and volume calculation logic.
- Connect Google Sheets nodes for database logging.
- Connect Gemini API using HTTP Request nodes.
- Configure Telegram response nodes.

4. Google Sheets Database Setup

- Create a Google Sheet named “Logs”.
- Add the following columns:
Date Exercise Weight Reps Sets Volume Category PR Status
- Connect Google Sheets credentials inside n8n.

5. Gemini AI Setup

- Generate a Gemini API key using Google AI Studio.
- Configure Gemini API requests inside n8n.
- Create prompts for workout coaching and progress summaries.

6. Workflow Testing

- Send voice workout messages through Telegram.
- Verify speech transcription accuracy.
- Check if workout data is logged correctly.
- Verify if AI coaching messages are generated.
- Test PR detection and progress summaries.

How It Works

1. The user sends a workout voice message through Telegram.
2. The Telegram Trigger node detects the incoming message.
3. The workflow retrieves and downloads the Telegram audio file.
4. The audio file is sent to the local Flask Whisper server.
5. Whisper AI transcribes the speech into text.

Example:

“inclined 24 kilogram 8 reps 3 sets”

1. JavaScript parsing logic extracts:
 - Exercise
 - Weight
 - Repetitions
 - Sets
2. The system categorizes the workout into:
 - Upper
 - Lower
 - Core



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3. The system calculates workout volume using:

$$\text{Volume} = \text{Weight} \times \text{Reps} \times \text{Sets}$$

1. The workflow checks historical workout logs inside Google Sheets.
2. The system detects whether the workout is a new Personal Record (PR).
3. The workout data is saved into Google Sheets.
4. Gemini AI analyzes the workout information.
5. Gemini generates motivational coaching feedback, progress analysis and summary.
6. The AI-generated response is sent back to the user through Telegram.



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Flow Chart

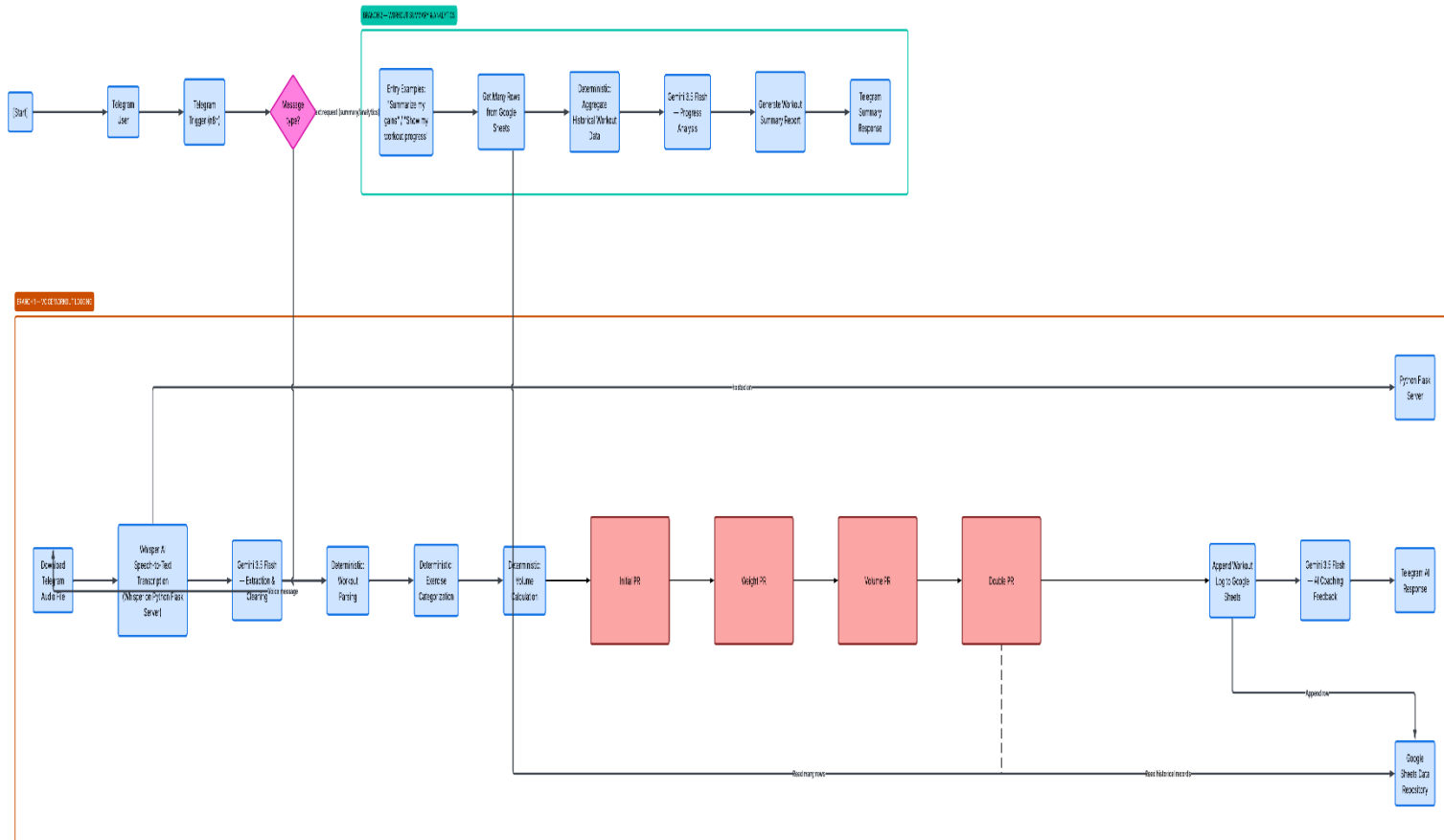


Figure 1 : generated from lucidchart

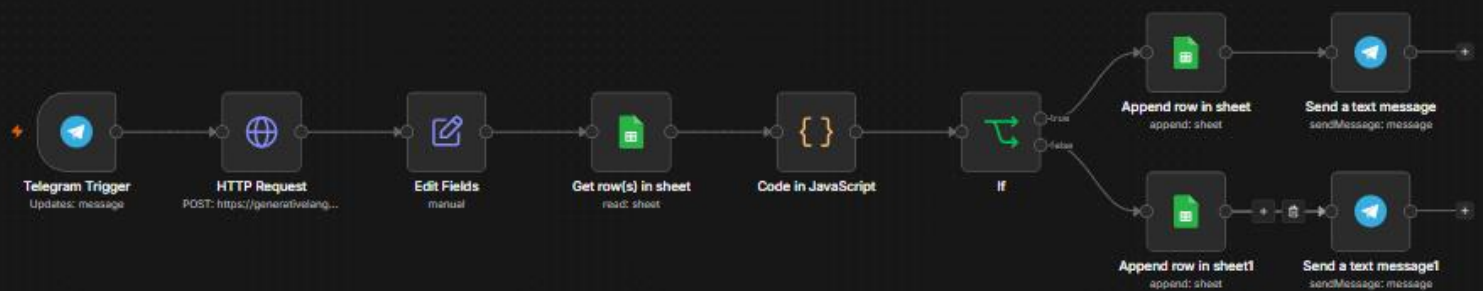


Figure 2 : actual version 1

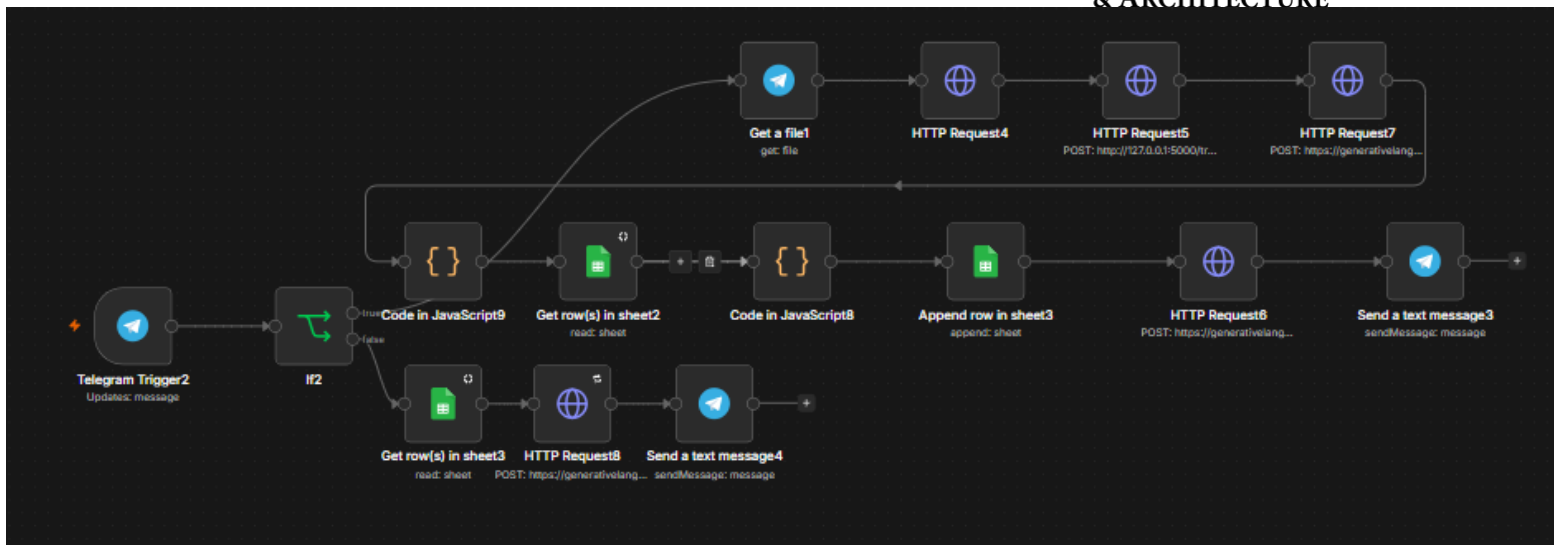


Figure 3: actual version 3

Results and Discussion

The Coach AI-rinx system successfully performed voice-based workout logging and AI-generated coaching. During testing, the system accurately transcribed workout voice messages using Whisper AI and extracted structured workout information such as exercise names, repetitions, sets, and workout weights.

The workflow successfully calculated workout volume and categorized workouts into different exercise groups such as upper body, lower body, arms, and core. The Google Sheets integration properly stored workout history and enabled the system to compare historical performance for personal record detection.

The Gemini AI integration also worked successfully by generating motivational coaching responses and workout summaries based on the logged workout data. The Telegram bot responded automatically with personalized feedback after each workout log.

The project demonstrated that artificial intelligence, automation workflows, and speech recognition technologies can be combined into a practical and intelligent fitness coaching system.

Conclusion

In conclusion, the Coach AI-rinx system successfully achieved its objectives by creating an AI-powered voice workout coaching and analytics platform. The project demonstrated how speech recognition, workflow automation, data analytics, and large language models can work together to automate workout tracking and provide intelligent coaching feedback.

The system reduced manual workout logging by allowing users to send voice messages instead of manually entering workout data. It also improved



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workout monitoring by automatically calculating analytics, detecting personal records, and generating motivational AI coaching responses.

This project proved that hybrid AI architectures combining deterministic logic and AI reasoning can create practical and user-friendly intelligent systems. The project also demonstrated the real-world application of modern AI technologies in health, fitness, and automation.

Therefore, Coach AI-rinx successfully serves as an AI-powered voice workout assistant and provides a strong foundation for future AI fitness coaching systems.

Future Improvements

Future versions of the project may include:

- Cloud deployment for 24/7 operation
- Mobile application integration
- Advanced workout recommendation systems
- Fatigue and recovery analysis
- AI-generated workout programs
- Dashboard analytics and charts
- Multi-user support
- Adaptive AI coaching memory

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